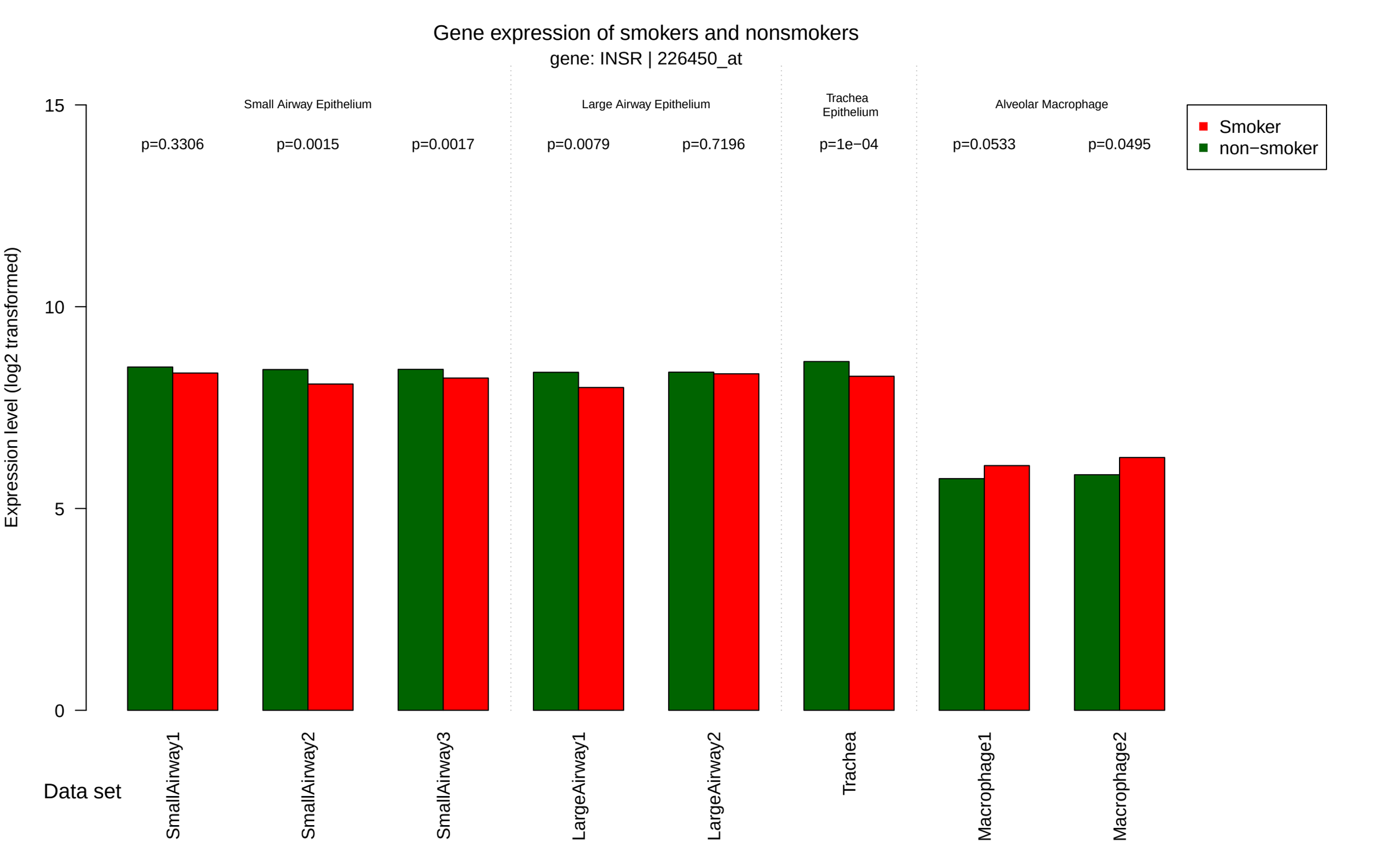
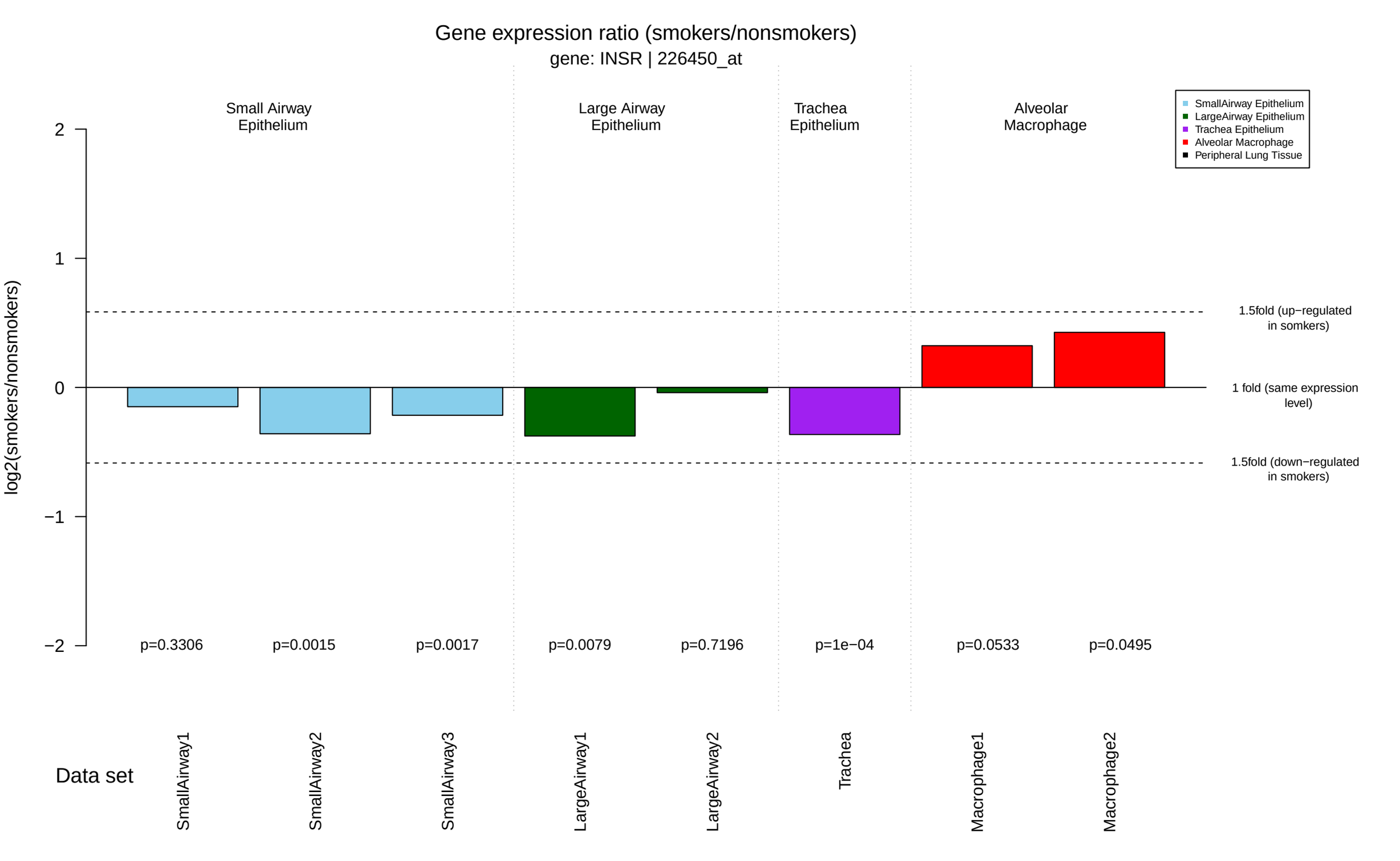
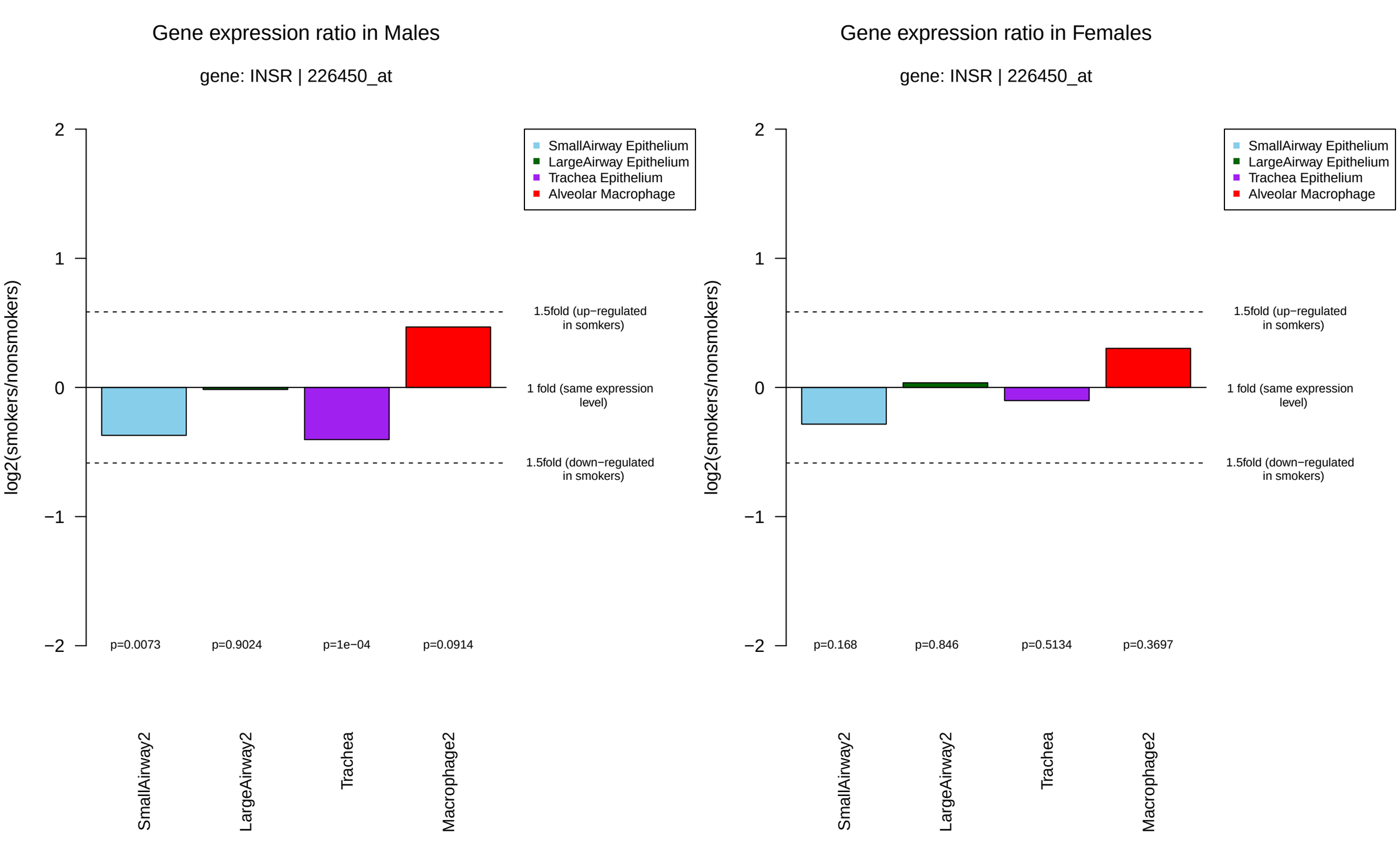
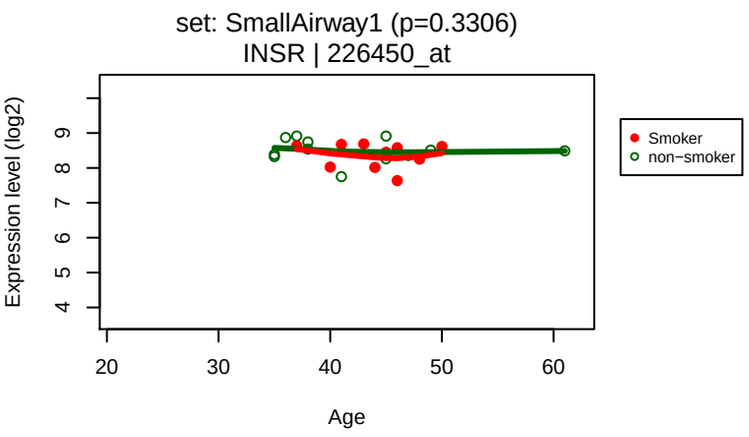
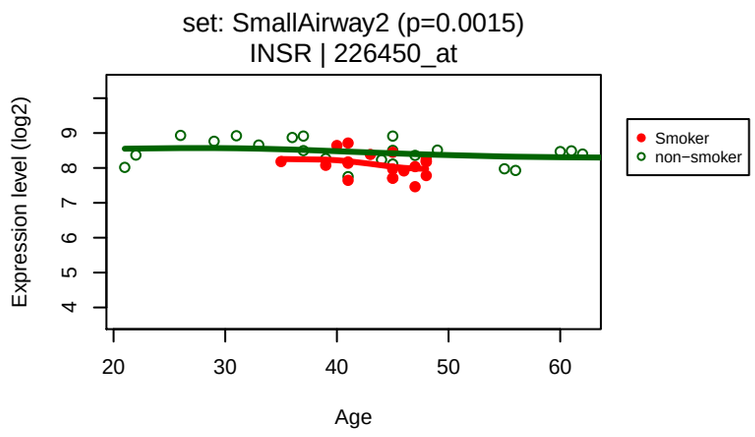
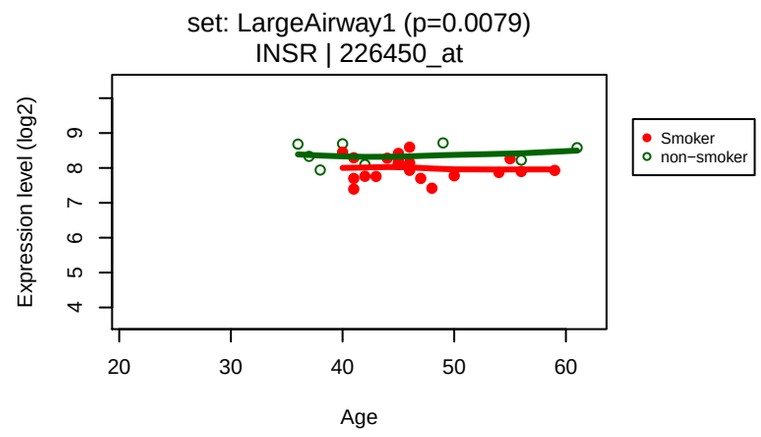

Gene expression of smokers and nonsmokers
gene: INSR | 226450_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3306
p=0.0015
p=0.0017
p=0.0079
p=0.7196
p=1e−04
p=0.0533
p=0.0495
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

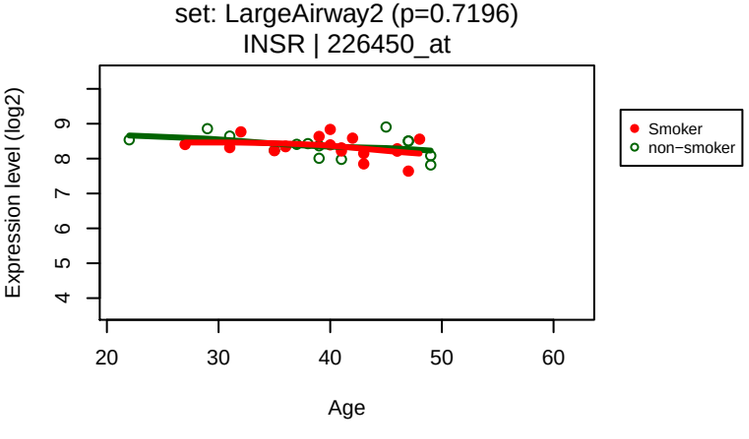

Gene expression ratio (smokers/nonsmokers)
gene: INSR | 226450_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3306
p=0.0015
p=0.0017
p=0.0079
p=0.7196
p=1e−04
p=0.0533
p=0.0495
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

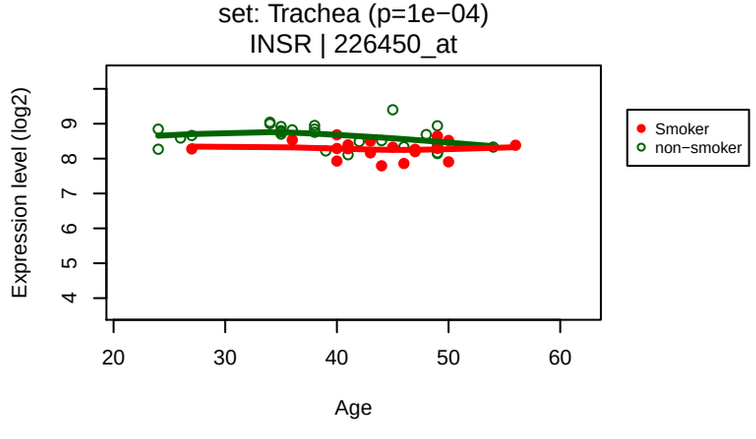

Gene expression ratio in Males
gene: INSR | 226450_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0073
p=0.9024
p=1e−04
p=0.0914
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: INSR | 226450_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.168
p=0.846
p=0.5134
p=0.3697
SmallAirway2
LargeAirway2
Trachea
Macrophage2

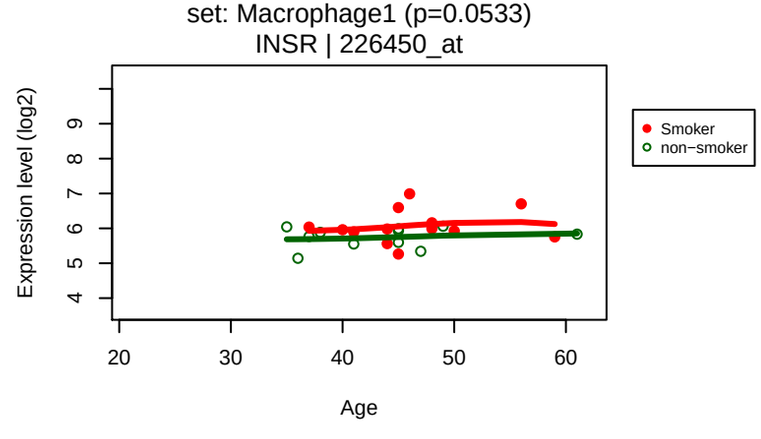

set: SmallAirway1 (p=0.3306)
INSR | 226450_at
Expression level (log2)
Age
Smoker
non−smoker

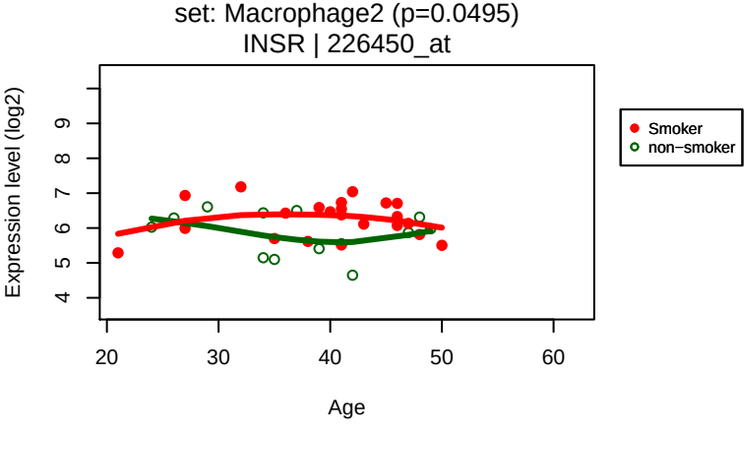

set: SmallAirway2 (p=0.0015)
INSR | 226450_at
Expression level (log2)
Age
Smoker
non−smoker

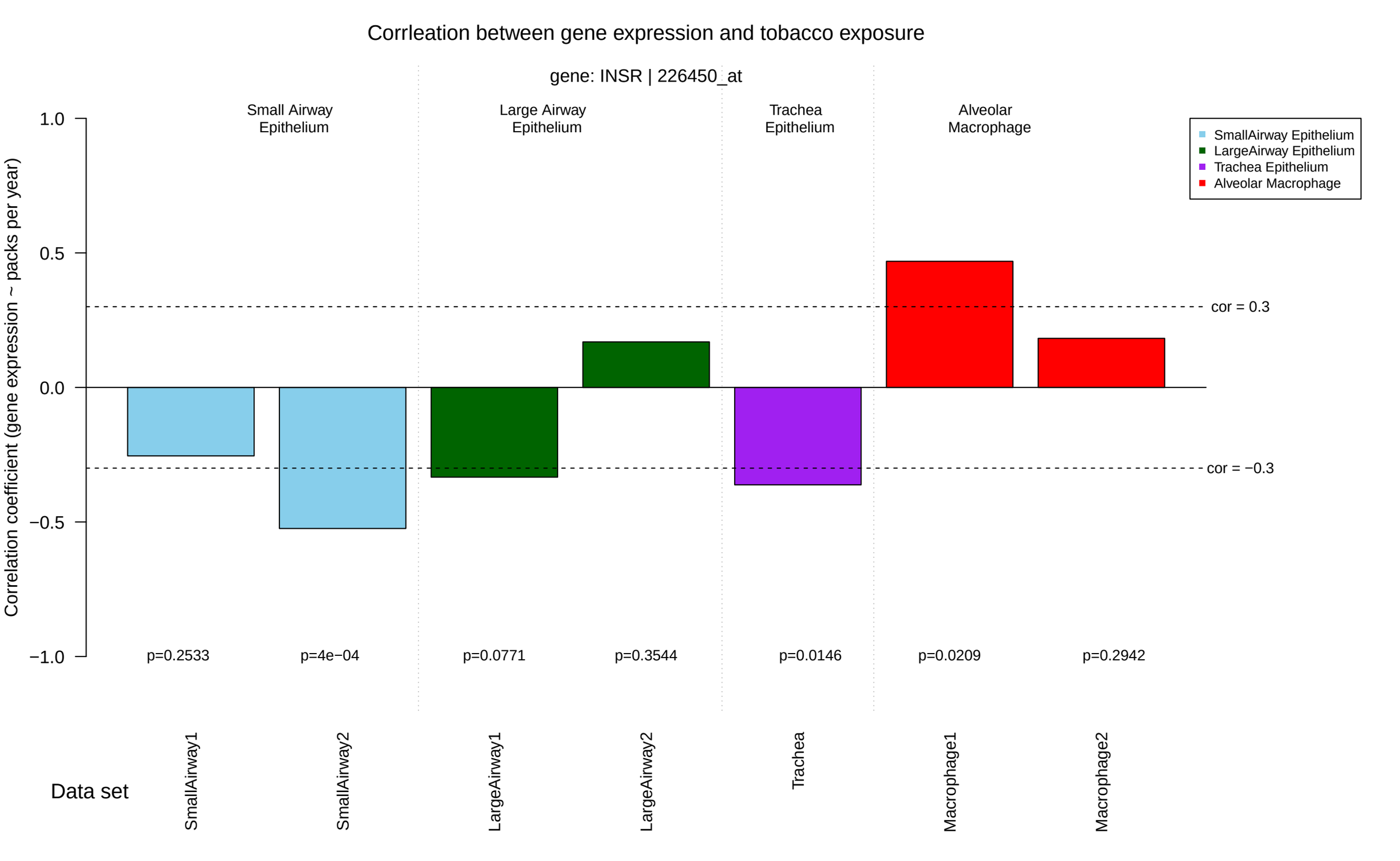

set: LargeAirway1 (p=0.0079)
INSR | 226450_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7196)
INSR | 226450_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
INSR | 226450_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0533)
INSR | 226450_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0495)
INSR | 226450_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: INSR | 226450_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2533
p=4e−04
p=0.0771
p=0.3544
p=0.0146
p=0.0209
p=0.2942
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2